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	AI Assisted Coding-Assignment:13 Date:15-10-2026

General Instructions

- Run the original program before making changes.
- Identify bugs systematically before asking AI for a complete solution.
- Use AI to explain causes and suggest fixes.
- Verify every AI suggestion by testing it yourself.
- Record the bug, cause, fix and test result.
- Do not blindly copy AI-generated code.

TASK 1 – Improve Readability

Set 1: Online Bakery Website

Story: A bakery owner asks an AI coding assistant such as Replit to create a responsive website to promote bakery products and receive customer enquiries.

Task:

1. Use AI to generate a responsive website using HTML, CSS, and JavaScript with the bakery name, product images, prices, and a contact form.
2. Use AI to create a REST API with GET /products to return bakery products in JSON format.
3. Test the website and API, and verify that the product details are displayed correctly.

PROMPT

- Create a responsive website using HTML, CSS, and JavaScript.
- Display the bakery name, product images, product names, prices, and descriptions.
- Add a customer contact/enquiry form with validation.
- Create a Python Flask REST API: GET `/products` that returns product details in JSON format.
- Implement POST `/contact` to process enquiries.
- Test the website and API.

CODE

SCRIPT.JS

```
const productGrid = document.querySelector('#product-grid');
const contactForm = document.querySelector('#contact-form');
const formStatus = document.querySelector('#form-status');

function createProductCard(product) {
```

```
  const card = document.createElement('article');
  card.className = 'product-card';

  const imageWrap = document.createElement('div');
  imageWrap.className = 'product-image-wrap';
  const image = document.createElement('img');
  image.className = 'product-image';
  image.src = product.image;
  image.alt = product.alt;
  image.loading = 'lazy';
  const category = document.createElement('span');
  category.className = 'product-category';
  category.textContent = product.category;
  imageWrap.append(image, category);

  const meta = document.createElement('div');
  meta.className = 'product-meta';
  const name = document.createElement('h3');
  name.textContent = product.name;
  const price = document.createElement('span');
  price.className = 'product-price';
  price.textContent = new Intl.NumberFormat('en-US', { style: 'currency',
currency: 'USD' }).format(product.price);
  meta.append(name, price);

  const description = document.createElement('p');
  description.className = 'product-description';
  description.textContent = product.description;
  card.append(imageWrap, meta, description);
  return card;
}

async function loadProducts() {
  try {
    const response = await fetch('/products');
    if (!response.ok) throw new Error('The menu is taking a little longer to
arrive.');
    const products = await response.json();
    productGrid.replaceChildren(...products.map(createProductCard));
  } catch (error) {
    const message = document.createElement('p');
    message.className = 'error-message';
```

```
    message.textContent = 'Today’s menu could not load. Please refresh the page
to try again.';
    productGrid.replaceChildren(message);
  }
}

contactForm.addEventListener('submit', async (event) => {
  event.preventDefault();
  formStatus.textContent = '';
  formStatus.classList.remove('is-error');

  if (!contactForm.reportValidity()) return;

  const submitButton = contactForm.querySelector('button[type="submit"]');
  submitButton.disabled = true;
  submitButton.setAttribute('aria-busy', 'true');
  submitButton.lastChild.textContent = ' Sending...';

  try {
    const response = await fetch('/contact', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(Object.fromEntries(new FormData(contactForm))),
    });
    const result = await response.json();
    if (!response.ok) throw new Error(result.error || 'Your note could not be
sent. Please try again.');
    formStatus.textContent = result.message;
    contactForm.reset();
  } catch (error) {
    formStatus.textContent = error.message || 'Something went wrong. Please try
again.';
    formStatus.classList.add('is-error');
  } finally {
    submitButton.disabled = false;
    submitButton.removeAttribute('aria-busy');
    submitButton.lastChild.textContent = ' Send your note ';
  }
});

loadProducts();
```

STYLE.CSS

```
@import
url('https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;600;700&family=
Fraunces:opsz,wght@9..144,500;9..144,600;9..144,700&display=swap');

:root {
  color-scheme: light;
  --paper: #f8f4e9;
  --paper-deep: #efe8d8;
  --ink: #292b22;
  --muted: #727267;
  --red: #c3432c;
  --yellow: #f1c94b;
  --green: #526848;
  --line: #ded8c9;
  --serif: 'Fraunces', Georgia, serif;
  --sans: 'DM Sans', sans-serif;
}

* { box-sizing: border-box; }
html { scroll-behavior: smooth; }
body { margin: 0; background: var(--paper); color: var(--ink); font-family: var(-
-sans); font-size: 15px; }
a { color: inherit; }
.announcement { min-height: 34px; padding: 8px 5.5%; display: flex; justify-
content: center; align-items: center; gap: 10px; background: var(--green); color:
#fffdf5; font-size: 11px; letter-spacing: .04em; text-align: center; }
.announcement span { color: var(--yellow); }
.site-header { height: 78px; padding: 0 5.5%; display: flex; align-items: center;
justify-content: space-between; border-bottom: 1px solid var(--line); }
.wordmark { font-family: var(--serif); font-size: 25px; font-weight: 700; text-
decoration: none; letter-spacing: -.04em; white-space: nowrap; }
.wordmark span { font-weight: 500; }
.wordmark i { color: var(--red); font-style: normal; }
.main-nav { display: flex; align-items: center; gap: 38px; }
.main-nav a, .header-note { font-size: 12px; text-decoration: none; }
.main-nav a:hover, .contact-details a:hover { color: var(--red); }
.header-note { padding: 10px 14px; border: 1px solid var(--line); border-radius:
2px; }
.header-note span { color: var(--red); margin-left: 9px; }
.intro { padding: 54px 5.5% 64px; display: grid; grid-template-columns: .83fr
1.17fr; gap: 6.5%; align-items: center; }
.intro-copy { padding: 9px 0 0 5%; }
```

```
.eyebrow { margin: 0 0 15px; color: var(--green); font-size: 10px; font-weight:
700; text-transform: uppercase; letter-spacing: .13em; }
.eyebrow-mark { color: var(--red); margin-right: 7px; }
h1, h2, p { margin-top: 0; }
h1, h2 { font-family: var(--serif); font-weight: 500; letter-spacing: -.035em; }
h1 { margin-bottom: 19px; font-size: clamp(48px, 5.5vw, 76px); line-height: .98;
}
h1 em, h2 em { color: var(--red); font-weight: 500; }
.intro-description { max-width: 370px; margin-bottom: 24px; color: #5f6057; font-
size: 14px; line-height: 1.8; }
.button { min-height: 46px; padding: 0 19px; display: inline-flex; gap: 25px;
align-items: center; justify-content: center; border: 0; border-radius: 2px;
color: white; font-family: var(--sans); font-size: 12px; font-weight: 600; text-
decoration: none; cursor: pointer; transition: transform .18s ease, background
.18s ease; }
.button:hover { transform: translateY(-2px); }
.button-dark { background: var(--ink); }
.button-dark:hover { background: var(--green); }
.button-red { background: var(--red); }
.button-red:hover { background: #a93420; }
.fresh-note { margin-top: 25px; color: var(--muted); font-size: 10px; }
.fresh-dot { width: 7px; height: 7px; margin-right: 8px; display: inline-block;
border-radius: 50%; background: #719356; }
.intro-photo { min-height: 405px; position: relative; }
.intro-photo > img { width: 100%; height: 430px; display: block; object-fit:
cover; object-position: center 55%; }
.photo-stamp { width: 116px; height: 116px; position: absolute; left: -28px;
bottom: 28px; display: flex; flex-direction: column; align-items: center;
justify-content: center; gap: 4px; border-radius: 50%; background: var(--yellow);
text-align: center; transform: rotate(-9deg); }
.photo-stamp span { font-size: 7px; font-weight: 700; letter-spacing: .12em; }
.photo-stamp strong { font-family: var(--serif); font-size: 18px; font-weight:
600; line-height: .95; }
.photo-caption { position: absolute; right: 0; bottom: -23px; color: var(--
muted); font-family: var(--serif); font-size: 12px; font-style: italic; }
.products-section { padding: 65px 5.5% 76px; border-top: 1px solid var(--line); }
.section-heading { margin-bottom: 30px; display: flex; align-items: end; justify-
content: space-between; gap: 20px; }
h2 { margin-bottom: 0; font-size: clamp(32px, 3.5vw, 46px); line-height: 1.1; }
.section-aside { margin-bottom: 4px; color: var(--muted); font-size: 11px; line-
height: 1.7; }
.product-grid { display: grid; grid-template-columns: repeat(3, minmax(0, 1fr));
gap: 34px 22px; }
.product-card { min-width: 0; }
```

```
.product-image-wrap { aspect-ratio: 1.28; overflow: hidden; position: relative;
background: var(--paper-deep); }
.product-image { width: 100%; height: 100%; display: block; object-fit: cover;
transition: transform .35s ease; }
.product-card:hover .product-image { transform: scale(1.035); }
.product-category { position: absolute; top: 12px; left: 12px; max-width:
calc(100% - 24px); padding: 7px 9px; background: var(--paper); color: var(--
green); font-size: 9px; font-weight: 700; letter-spacing: .06em; text-transform:
uppercase; }
.product-meta { margin-top: 13px; display: flex; justify-content: space-between;
gap: 12px; align-items: baseline; }
.product-meta h3 { margin: 0; font-family: var(--serif); font-size: 19px; font-
weight: 600; line-height: 1.2; }
.product-price { flex: 0 0 auto; color: var(--red); font-size: 13px; font-weight:
700; }
.product-description { max-width: 390px; margin: 7px 0 0; color: var(--muted);
font-size: 11px; line-height: 1.65; }
.loading-message, .error-message { grid-column: 1 / -1; padding: 28px 0; color:
var(--muted); }
.story-strip { padding: 48px 8%; display: grid; grid-template-columns: 100px 1fr
1fr; gap: 28px; align-items: center; background: var(--paper-deep); }
.story-mark { width: 76px; height: 76px; display: grid; place-items: center;
border: 1px solid #b8ad94; border-radius: 50%; color: var(--red); font-family:
var(--serif); font-size: 29px; }
.story-mark span { margin-left: -25px; margin-top: 22px; color: var(--green); }
.story-strip h2 { font-size: 34px; }
.story-copy { max-width: 390px; margin: 0; color: #5f6057; font-size: 12px; line-
height: 1.8; }
.contact-section { padding: 75px 11%; display: grid; grid-template-columns: 1fr
1fr; gap: 12%; }
.contact-copy h2 { margin-bottom: 16px; font-size: 48px; }
.contact-copy > p:not(.eyebrow) { max-width: 340px; color: var(--muted); font-
size: 13px; line-height: 1.75; }
.contact-details { margin-top: 30px; display: grid; gap: 9px; font-size: 11px; }
.contact-details a { font-weight: 700; text-decoration-color: var(--red); text-
underline-offset: 3px; }
.contact-details span { color: var(--muted); }
.contact-form { display: grid; grid-template-columns: 1fr; }
.contact-form label { margin: 0 0 7px; font-size: 11px; font-weight: 700; }
.contact-form input, .contact-form textarea { width: 100%; margin-bottom: 16px;
padding: 12px 13px; border: 1px solid var(--line); border-radius: 2px; outline:
none; background: #fffdf7; color: var(--ink); font: 12px var(--sans); }
.contact-form input:focus, .contact-form textarea:focus { border-color: var(--
green); box-shadow: 0 0 0 2px #52684820; }
```

```
.contact-form textarea { min-height: 105px; resize: vertical; }
.form-footer { min-height: 46px; display: flex; justify-content: space-between;
align-items: center; gap: 12px; }
.form-status { margin: 0; color: var(--green); font-size: 11px; line-height: 1.5;
}
.form-status.is-error { color: var(--red); }
.form-footer .button { flex: 0 0 auto; }
.site-footer { min-height: 76px; padding: 16px 5.5%; display: flex; justify-
content: space-between; align-items: center; gap: 18px; border-top: 1px solid
var(--line); }
.site-footer .wordmark { font-size: 20px; }
.site-footer > span { color: var(--muted); font-size: 10px; }

@media (max-width: 850px) {
  .main-nav { gap: 20px; }
  .header-note { display: none; }
  .intro { grid-template-columns: .9fr 1.1fr; gap: 5%; }
  .intro-copy { padding-left: 0; }
  .intro-photo, .intro-photo > img { min-height: 350px; height: 350px; }
  .story-strip { padding: 40px 5.5%; grid-template-columns: 80px 1fr 1fr; gap:
18px; }
  .story-strip h2 { font-size: 29px; }
  .contact-section { padding-right: 7%; padding-left: 7%; gap: 8%; }
}

@media (max-width: 620px) {
  .announcement { padding-right: 12px; padding-left: 12px; font-size: 9px; }
  .site-header { height: auto; min-height: 70px; padding: 13px 5%; align-items:
flex-start; flex-direction: column; gap: 10px; }
  .wordmark { font-size: 23px; }
  .main-nav { width: 100%; justify-content: space-between; gap: 12px; }
  .main-nav a { font-size: 10px; }
  .intro { padding: 40px 5% 50px; grid-template-columns: 1fr; gap: 28px; }
  .intro-copy { padding: 0; }
  h1 { font-size: 58px; }
  .intro-description { max-width: 410px; }
  .intro-photo, .intro-photo > img { min-height: 290px; height: 290px; }
  .photo-stamp { width: 92px; height: 92px; left: -9px; bottom: 22px; }
  .photo-stamp strong { font-size: 15px; }
  .photo-stamp span { font-size: 6px; }
  .products-section { padding: 48px 5% 55px; }
  .section-heading { align-items: flex-start; flex-direction: column; gap: 12px;
}
  .section-aside br { display: none; }
```

```
  .product-grid { grid-template-columns: repeat(2, minmax(0, 1fr)); gap: 26px
13px; }
  .product-image-wrap { aspect-ratio: .95; }
  .product-category { top: 8px; left: 8px; padding: 6px 7px; font-size: 7px; }
  .product-meta { margin-top: 9px; align-items: flex-start; flex-direction:
column; gap: 3px; }
  .product-meta h3 { font-size: 17px; }
  .product-price { font-size: 12px; }
  .product-description { font-size: 10px; }
  .story-strip { padding: 36px 6%; grid-template-columns: 60px 1fr; gap: 16px; }
  .story-mark { width: 54px; height: 54px; font-size: 23px; }
  .story-mark span { margin-left: -20px; margin-top: 18px; }
  .story-strip h2 { font-size: 27px; }
  .story-copy { grid-column: 2; }
  .contact-section { padding: 54px 6%; grid-template-columns: 1fr; gap: 30px; }
  .contact-copy h2 { font-size: 42px; }
  .site-footer { padding: 19px 5%; flex-wrap: wrap; }
  .site-footer > span { font-size: 9px; }
}

@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after { scroll-behavior: auto !important; transition-duration:
.01ms !important; }
}
```

INDEX.HTML

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <meta name="theme-color" content="#f8f4e9">
  <meta name="description" content="Small-batch breads and pastries, baked fresh
every morning at Sunday Crumb.">
  <title>Sunday Crumb | Good mornings, made by hand</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
href="https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;600;700&family
=Fraunces:opsz,wght@9..144,500;9..144,600;9..144,700&display=swap"
rel="stylesheet">
  <link rel="stylesheet" href="{{ url_for('static', filename='styles.css') }}">
  <script src="{{ url_for('static', filename='script.js') }}" defer></script>
```

```
</head>
<body>
  <div class="announcement">Fresh from the oven, every day from 7am <span aria-hidden="true">✳</span> 14 Orchard Lane, Brooklyn</div>
  <header class="site-header">
    <a class="wordmark" href="#home" aria-label="Sunday Crumb home">sunday<span>crumb</span><i aria-hidden="true">.</i></a>
    <nav class="main-nav" aria-label="Main navigation">
      <a href="#bakes">The bakes</a>
      <a href="#story">Our little bakery</a>
      <a href="#contact">Say hello</a>
    </nav>
    <a class="header-note" href="#bakes">Baked with a little extra love <span aria-hidden="true">↗</span></a>
  </header>

  <main id="home">
    <section class="intro" aria-labelledby="intro-title">
      <div class="intro-copy">
        <p class="eyebrow"><span class="eyebrow-mark" aria-hidden="true">✳</span> Your neighborhood bakery</p>
        <h1 id="intro-title">A good day<br>starts <em>with crumbs.</em></h1>
        <p class="intro-description">Slow mornings, warm bread, one more pastry than you planned. We make the good stuff by hand, right here in Brooklyn.</p>
        <a class="button button-dark" href="#bakes">Find your favorite <span aria-hidden="true">↓</span></a>
        <div class="fresh-note"><span class="fresh-dot"></span> Out of the oven daily, while it lasts</div>
      </div>
      <div class="intro-photo">
        <img src="https://images.unsplash.com/photo-1555507036-ab1f4038808a?auto=format&fit=crop&w=1600&q=90" alt="Freshly baked croissants on a tray" fetchpriority="high">
        <div class="photo-stamp" aria-label="Made fresh every day"><span>MADE FRESH</span><strong>every<br>single day</strong><span>♡   SINCE 2018</span></div>
        <span class="photo-caption">A little butter makes everything better.</span>
      </div>
    </section>

    <section class="products-section" id="bakes" aria-labelledby="products-title">
```

```
      <div class="section-heading">
        <div><p class="eyebrow">The morning line-up</p><h2 id="products-
title">Good things, <em>fresh daily.</em></h2></div>
        <p class="section-aside">Small batches, seasonal ingredients,<br>and
absolutely no rushing the dough.</p>
      </div>
      <div class="product-grid" id="product-grid" aria-live="polite">
        <p class="loading-message">Pulling today's favorites from the oven...</p>
      </div>
    </section>

    <section class="story-strip" id="story" aria-labelledby="story-title">
      <div class="story-mark" aria-hidden="true">S<span>C</span></div>
      <div><p class="eyebrow">A small note from us</p><h2 id="story-title">Good
bread takes time.<br><em>So does a good morning.</em></h2></div>
      <p class="story-copy">We're a tiny neighborhood bakery that believes in
real butter, long fermentation, and knowing your regulars by name. Come as you
are. Leave with something warm.</p>
    </section>

    <section class="contact-section" id="contact" aria-labelledby="contact-
title">
      <div class="contact-copy"><p class="eyebrow">Big order? Little
question?</p><h2 id="contact-title">We're all <em>ears.</em></h2><p>Tell us what
you're dreaming up. We'll get back to you within one bakery day.</p><div
class="contact-details"><a
href="mailto:hello@sundaycrumb.com">hello@sundaycrumb.com</a><span>14 Orchard
Lane, Brooklyn, NY</span></div></div>
      <form class="contact-form" id="contact-form" novalidate>
        <label for="name">Your name</label>
        <input id="name" name="name" autocomplete="name" required minlength="2"
maxlength="80" placeholder="The name we should use">
        <label for="email">Email address</label>
        <input id="email" name="email" type="email" autocomplete="email" required
placeholder="you@example.com">
        <label for="message">What's on your mind?</label>
        <textarea id="message" name="message" rows="4" minlength="10"
maxlength="2000" required placeholder="A custom cake, a big order, or just
hello..."></textarea>
        <div class="form-footer"><p class="form-status" id="form-status"
role="status" aria-live="polite"></p><button class="button button-red"
type="submit">Send your note <span aria-hidden="true">↗</span></button></div>
      </form>
```

```
    </section>
  </main>

  <footer class="site-footer"><a class="wordmark"
href="#home">sunday<span>crumb</span><i aria-hidden="true">.</i></a><span>Made
with butter and good intentions.</span><span>Open daily, 7am–3pm</span></footer>
</body>
</html>
```

MAIN.PY

```
from flask import Flask, jsonify, render_template, request
import re

app = Flask(__name__)

PRODUCTS = [
    {
        "id": 1,
        "name": "Butter Croissant",
        "category": "Morning favorite",
        "price": 4.25,
        "description": "Laminated by hand, baked until deeply golden, and
finished with cultured butter.",
        "image": "https://images.unsplash.com/photo-1555507036-
ab1f4038808a?auto=format&fit=crop&w=900&q=85",
        "alt": "Golden, flaky butter croissants",
    },
    {
        "id": 2,
        "name": "Wildflower Honey Loaf",
        "category": "Naturally leavened",
        "price": 9.50,
        "description": "A slow-fermented country loaf with a tender crumb and a
whisper of local honey.",
        "image": "https://images.unsplash.com/photo-1509440159596-
0249088772ff?auto=format&fit=crop&w=900&q=85",
        "alt": "Freshly baked artisan bread loaf",
    },
    {
        "id": 3,
        "name": "Raspberry Morning Bun",
        "category": "Small-batch pastry",
```

```
        "price": 5.75,
        "description": "Crisp, cinnamon-sugar layers with bright raspberry jam tucked inside.",
        "image": "https://images.unsplash.com/photo-1509365465985-25d11c17e812?auto=format&fit=crop&w=900&q=85",
        "alt": "Fresh pastries displayed in a bakery",
    },
    {
        "id": 4,
        "name": "Olive & Sea Salt Focaccia",
        "category": "Baked this morning",
        "price": 8.00,
        "description": "Olive-oil-rich, pillowy focaccia topped with briny olives and flaky sea salt.",
        "image": "https://images.unsplash.com/photo-1573140247632-f8fd74997d5c?auto=format&fit=crop&w=900&q=85",
        "alt": "Rustic loaf of bakery bread",
    },
    {
        "id": 5,
        "name": "Dark Chocolate Cookie",
        "category": "A little treat",
        "price": 3.50,
        "description": "Chewy-centered and crisp at the edges, with generous bittersweet chocolate.",
        "image": "https://images.unsplash.com/photo-1499636136210-6f4ee915583e?auto=format&fit=crop&w=900&q=85",
        "alt": "Chocolate chip cookies fresh from the oven",
    },
    {
        "id": 6,
        "name": "Lemon Ricotta Cake",
        "category": "Weekend special",
        "price": 6.25,
        "description": "Soft ricotta cake brightened with fresh lemon zest and a little powdered sugar.",
        "image": "https://images.unsplash.com/photo-1578985545062-69928b1d9587?auto=format&fit=crop&w=900&q=85",
        "alt": "A beautifully finished bakery cake",
    },
]
```

```
# Enquiries are kept for the lifetime of this process; replace with persistent
storage for deployment.
CONTACT_ENQUIRIES = []
EMAIL_PATTERN = re.compile(r"^[^\s@]+@[^\s@]+\.[^\s@]+$")



@app.get("/")
def home():
    return render_template("index.html")



@app.get("/products")
def get_products():
    return jsonify(PRODUCTS)



@app.post("/contact")
def create_contact_enquiry():
    payload = request.get_json(silent=True)
    if not isinstance(payload, dict):
        return jsonify({"error": "Please send your enquiry as JSON."}), 400

    name = payload.get("name", "")
    email = payload.get("email", "")
    message = payload.get("message", "")
    if not all(isinstance(value, str) for value in (name, email, message)):
        return jsonify({"error": "Name, email, and message must be text."}), 400

    name, email, message = name.strip(), email.strip(), message.strip()
    if len(name) < 2 or len(name) > 80:
        return jsonify({"error": "Please enter a name between 2 and 80
characters."}), 400
    if not EMAIL_PATTERN.fullmatch(email):
        return jsonify({"error": "Please enter a valid email address."}), 400
    if len(message) < 10 or len(message) > 2000:
        return jsonify({"error": "Your message must be between 10 and 2000
characters."}), 400

    enquiry = {"name": name, "email": email, "message": message}
    CONTACT_ENQUIRIES.append(enquiry)
    app.logger.info("New bakery enquiry from %s <%s>", name, email)
```

```
    return jsonify({"message": "Thanks for reaching out. We will be in touch
soon."}), 201


if __name__ == "__main__":
    app.run(debug=True)
```

TEST_APP.PY

```
import unittest

from app import app, CONTACT_ENQUIRIES, PRODUCTS


class BakeryApiTests(unittest.TestCase):
    def setUp(self):
        self.client = app.test_client()
        CONTACT_ENQUIRIES.clear()

    def test_homepage_is_served(self):
        response = self.client.get("/")
        self.assertEqual(response.status_code, 200)
        self.assertIn(b"Sunday Crumb", response.data)

    def test_products_returns_catalog(self):
        response = self.client.get("/products")
        self.assertEqual(response.status_code, 200)
        products = response.get_json()
        self.assertEqual(len(products), len(PRODUCTS))
        self.assertTrue(all({"id", "name", "price", "description", "image"} <=
product.keys() for product in products))

    def test_contact_accepts_valid_enquiry(self):
        response = self.client.post("/contact", json={
            "name": "Alex Baker",
            "email": "alex@example.com",
            "message": "Could you make a birthday cake for Saturday?",
        })
        self.assertEqual(response.status_code, 201)
        self.assertEqual(len(CONTACT_ENQUIRIES), 1)

    def test_contact_rejects_invalid_email(self):
```

```
        response = self.client.post("/contact", json={
            "name": "Alex Baker",
            "email": "not-an-email",
            "message": "Could you make a birthday cake for Saturday?",
        })
        self.assertEqual(response.status_code, 400)
        self.assertEqual(len(CONTACT_ENQUIRIES), 0)

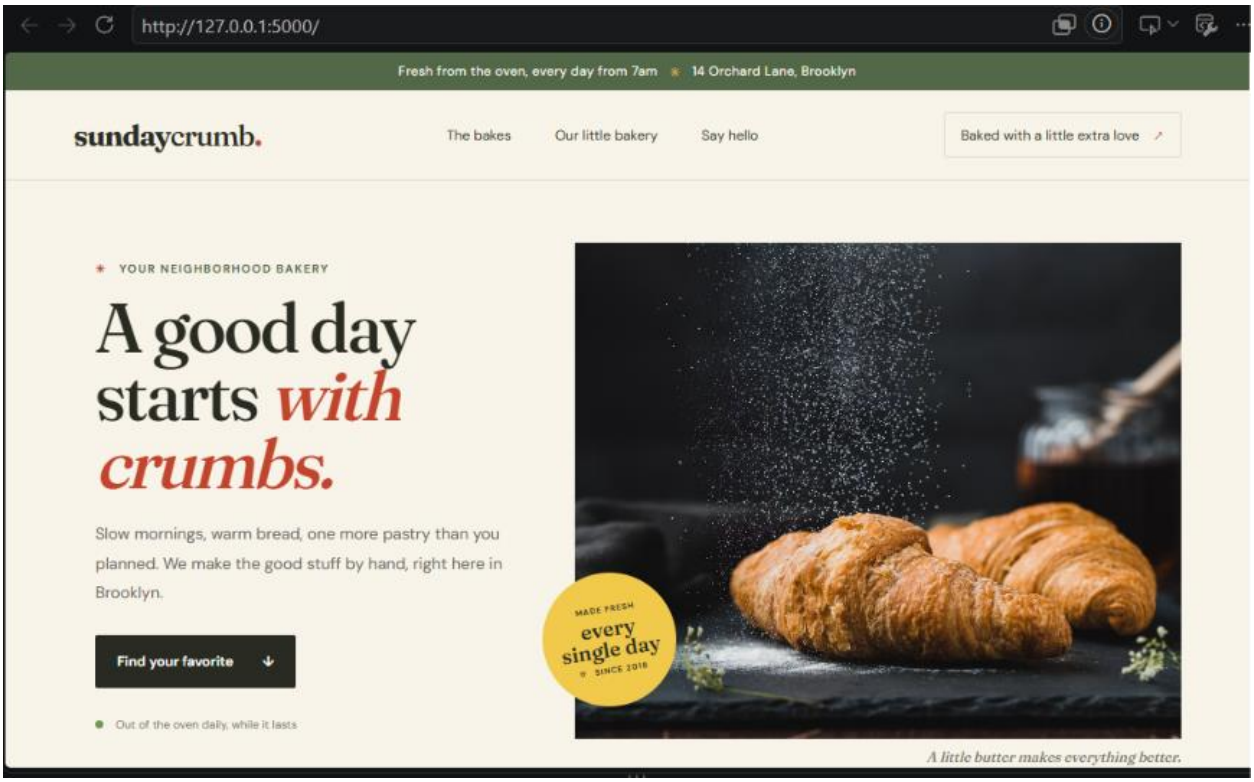
    def test_contact_rejects_non_json_request(self):
        response = self.client.post("/contact", data="hello")
        self.assertEqual(response.status_code, 400)


if __name__ == "__main__":
    unittest.main()
```

REQUIREMENTS.TXT

```
Flask>=3.0,<4.0
```

OUTPUT



Set 2: Student Course Registration Website

Story: A college wants a simple website where students can view available courses and submit registration details.

Task:

1. Use AI to generate a responsive frontend with a course list, registration form, and JavaScript form validation.
2. Use AI to develop REST API endpoints: GET /courses to retrieve courses and POST /registrations to submit student registrations.
3. Test the frontend and API to verify course retrieval and registration submission.

PROMPT

- Create a responsive frontend with a course list and student registration form.
- Add JavaScript form validation.
- Create Flask REST API endpoints:
  - `GET /courses` — retrieve available courses.
  - `POST /registrations` — submit student registration details.
- Validate required fields and course selection.
- Test course retrieval and registration submission.

CODE

ACADAMY.CSS

```
@import
url('https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;600;700&family=
Fraunces:opsz,wght@9..144,500;9..144,600;9..144,700&display=swap');

:root {
  color-scheme: light;
  --paper: #f4f5ee;
  --white: #fffef9;
  --ink: #202d29;
  --muted: #707b73;
```

```
  --green: #31594b;
  --lime: #d6ed68;
  --blue: #c5d7e9;
  --coral: #e5785d;
  --line: #d9ded4;
  --serif: 'Fraunces', Georgia, serif;
  --sans: 'DM Sans', sans-serif;
}

* { box-sizing: border-box; }
html { scroll-behavior: smooth; }
body { margin: 0; background: var(--paper); color: var(--ink); font-family: var(-
-sans); font-size: 15px; }
a { color: inherit; }
.topline { min-height: 34px; padding: 8px 5.5%; display: flex; justify-content:
space-between; gap: 15px; background: var(--green); color: #f4f5ee; font-size:
9px; letter-spacing: .07em; }
.topline > span:first-child { font-weight: 700; }
.topline b { margin: 0 8px; color: var(--lime); }
.site-header { min-height: 78px; padding: 12px 5.5%; display: flex; justify-
content: space-between; align-items: center; border-bottom: 1px solid var(--
line); }
.brand { display: inline-flex; gap: 10px; align-items: center; text-decoration:
none; font-family: var(--serif); font-size: 21px; font-weight: 600; line-height:
.9; }
.brand-symbol { width: 35px; height: 35px; display: grid; place-items: center;
border-radius: 50%; background: var(--lime); color: var(--ink); font-size: 24px;
}
.brand small { margin-top: 5px; display: block; font-family: var(--sans); font-
size: 7px; font-weight: 700; letter-spacing: .16em; }
.navigation { display: flex; align-items: center; gap: 32px; }
.navigation a { font-size: 11px; text-decoration: none; }
.navigation a:hover, .footer > a:last-child:hover { color: var(--coral); }
.navigation .nav-cta { padding: 11px 14px; border: 1px solid var(--line); border-
radius: 2px; }
.nav-cta span { margin-left: 8px; color: var(--coral); }
.hero { padding: 52px 5.5% 67px; display: grid; grid-template-columns: .83fr
1.17fr; gap: 6.5%; align-items: center; }
.hero-copy { padding-left: 5%; }
.eyebrow { margin: 0 0 15px; color: var(--green); font-size: 9px; font-weight:
700; letter-spacing: .14em; text-transform: uppercase; }
.eyebrow-line { width: 22px; height: 1px; margin-right: 8px; display: inline-
block; vertical-align: middle; background: var(--coral); }
```

```
h1, h2 { margin: 0; font-family: var(--serif); font-weight: 500; letter-spacing:
-.035em; }
h1 { font-size: clamp(48px, 5.2vw, 73px); line-height: .99; }
h1 em, h2 em { color: var(--green); font-weight: 500; }
.hero-description { max-width: 365px; margin: 20px 0 23px; color: #5f6c64; font-
size: 13px; line-height: 1.8; }
.hero-actions { display: flex; align-items: center; gap: 19px; flex-wrap: wrap; }
.button { min-height: 44px; padding: 0 17px; display: inline-flex; justify-
content: center; align-items: center; gap: 24px; border: 0; border-radius: 2px;
color: white; font-family: var(--sans); font-size: 11px; font-weight: 700; text-
decoration: none; cursor: pointer; transition: transform .18s ease, background
.18s ease; }
.button:hover { transform: translateY(-2px); }
.button-ink { background: var(--ink); }
.button-ink:hover { background: var(--green); }
.button-coral { background: var(--coral); }
.button-coral:hover { background: #ca5e45; }
.class-note { color: var(--muted); font-size: 10px; }
.class-note b { margin-right: 5px; color: var(--coral); font-size: 15px; }
.hero-footnote { margin-top: 31px; color: var(--muted); font-family: var(--
serif); font-size: 12px; font-style: italic; }
.hero-visual { min-height: 390px; position: relative; }
.hero-visual > img { width: 100%; height: 415px; display: block; object-fit:
cover; object-position: center; }
.hero-sticker { width: 112px; height: 112px; position: absolute; left: -27px;
bottom: 24px; display: flex; flex-direction: column; justify-content: center;
align-items: center; gap: 5px; border-radius: 50%; background: var(--lime); text-
align: center; transform: rotate(-8deg); }
.hero-sticker span { font-size: 7px; font-weight: 700; letter-spacing: .11em; }
.hero-sticker strong { font-family: var(--serif); font-size: 17px; line-height:
.95; }
.image-caption { position: absolute; right: 0; bottom: -22px; color: var(--
muted); font-family: var(--serif); font-size: 11px; font-style: italic; }
.course-section { padding: 64px 5.5% 75px; border-top: 1px solid var(--line); }
.section-heading { margin-bottom: 30px; display: flex; justify-content: space-
between; align-items: end; gap: 20px; }
h2 { font-size: clamp(32px, 3.3vw, 44px); line-height: 1.1; }
.section-heading > p { margin: 0 0 3px; color: var(--muted); font-size: 10px;
line-height: 1.7; }
.course-grid { display: grid; grid-template-columns: repeat(3, minmax(0, 1fr));
gap: 22px; }
.course-card { min-width: 0; background: var(--white); }
.course-image-frame { aspect-ratio: 1.37; position: relative; overflow: hidden;
background: var(--blue); }
```

```
.course-lime .course-image-frame { background: var(--lime); }
.course-blue .course-image-frame { background: var(--blue); }
.course-coral .course-image-frame { background: #f1c8b9; }
.course-image-frame img { width: 100%; height: 100%; display: block; object-fit: cover; transition: transform .35s ease; }
.course-card:hover .course-image-frame img { transform: scale(1.035); }
.course-number { width: 35px; height: 35px; position: absolute; top: 11px; left: 11px; display: grid; place-items: center; border-radius: 50%; background: var(--white); font-family: var(--serif); font-size: 13px; }
.course-details { padding: 17px 17px 14px; }
.course-category { margin: 0 0 7px; color: var(--green); font-size: 8px; font-weight: 700; letter-spacing: .11em; text-transform: uppercase; }
.course-details h3 { min-height: 44px; margin: 0; font-family: var(--serif); font-size: 20px; font-weight: 600; line-height: 1.15; }
.course-description { min-height: 51px; margin: 9px 0 14px; color: var(--muted); font-size: 10px; line-height: 1.65; }
.course-meta { padding-top: 11px; display: flex; justify-content: space-between; gap: 8px; border-top: 1px solid var(--line); color: var(--muted); font-size: 9px; }
.course-choose { width: 100%; margin-top: 12px; padding: 9px 0 2px; display: flex; justify-content: space-between; border: 0; background: transparent; color: var(--ink); font: 600 10px var(--sans); text-align: left; cursor: pointer; }
.course-choose::after { content: '↗'; color: var(--coral); }
.course-choose:hover { color: var(--green); }
.loading-message, .error-message { grid-column: 1 / -1; padding: 26px 0; color: var(--muted); }
.approach { padding: 46px 8%; display: grid; grid-template-columns: 80px 1fr 1fr; gap: 30px; align-items: center; background: #e6ebdf; }
.approach-index { align-self: start; padding-top: 6px; color: var(--coral); font-family: var(--serif); font-size: 17px; }
.approach-index span { color: var(--muted); font-family: var(--sans); font-size: 9px; }
.approach h2 { font-size: 34px; }
.approach-copy { max-width: 390px; margin: 0; color: #5f6c64; font-size: 11px; line-height: 1.8; }
.registration { padding: 75px 11%; display: grid; grid-template-columns: 1fr 1fr; gap: 12%; }
.registration-copy h2 { margin-bottom: 15px; font-size: 47px; }
.registration-copy > p:not(.eyebrow) { max-width: 350px; color: var(--muted); font-size: 12px; line-height: 1.8; }
.registration-aside { max-width: 270px; margin-top: 30px; padding-top: 13px; display: flex; align-items: center; gap: 10px; border-top: 1px solid var(--line); color: var(--green); font-size: 10px; }
.aside-star { color: var(--coral); font-size: 18px; }
```

```
.registration-form { display: grid; grid-template-columns: 1fr; }
.registration-form label { margin: 0 0 7px; font-size: 10px; font-weight: 700; }
.registration-form input, .registration-form select { width: 100%; min-height:
43px; margin-bottom: 16px; padding: 0 12px; border: 1px solid var(--line);
border-radius: 2px; outline: none; background: var(--white); color: var(--ink);
font: 11px var(--sans); }
.registration-form input:focus, .registration-form select:focus { border-color:
var(--green); box-shadow: 0 0 0 2px #31594b20; }
.form-bottom { min-height: 45px; display: flex; justify-content: space-between;
align-items: center; gap: 12px; }
.form-status { margin: 0; color: var(--green); font-size: 10px; line-height: 1.5;
}
.form-status.is-error { color: #b3402d; }
.form-bottom .button { flex: 0 0 auto; }
button:disabled { cursor: wait; opacity: .65; }
.footer { min-height: 75px; padding: 16px 5.5%; display: flex; justify-content:
space-between; align-items: center; gap: 18px; border-top: 1px solid var(--line);
}
.footer .brand { font-size: 18px; }
.footer .brand-symbol { width: 30px; height: 30px; font-size: 20px; }
.footer > span, .footer > a:last-child { color: var(--muted); font-size: 9px; }

@media (max-width: 850px) {
  .navigation { gap: 18px; }
  .hero { grid-template-columns: .9fr 1.1fr; gap: 5%; }
  .hero-copy { padding-left: 0; }
  .hero-visual, .hero-visual > img { min-height: 345px; height: 345px; }
  .approach { padding: 40px 5.5%; grid-template-columns: 58px 1fr 1fr; gap: 20px;
}
  .approach h2 { font-size: 29px; }
  .registration { padding-right: 7%; padding-left: 7%; gap: 8%; }
}

@media (max-width: 620px) {
  .topline { padding: 8px 5%; font-size: 8px; }
  .topline > span:first-child { display: none; }
  .topline > span:last-child { margin: auto; text-align: center; }
  .site-header { min-height: 70px; padding: 12px 5%; align-items: flex-start;
flex-direction: column; gap: 12px; }
  .navigation { width: 100%; justify-content: space-between; gap: 10px; }
  .navigation a { font-size: 9px; }
  .navigation .nav-cta { padding: 8px; }
  .hero { padding: 39px 5% 49px; grid-template-columns: 1fr; gap: 27px; }
  .hero-copy { padding: 0; }
```

```
  h1 { font-size: 57px; }
  .hero-description { max-width: 420px; }
  .hero-visual, .hero-visual > img { min-height: 285px; height: 285px; }
  .hero-sticker { width: 92px; height: 92px; left: -8px; bottom: 20px; }
  .hero-sticker strong { font-size: 15px; }
  .hero-sticker span { font-size: 6px; }
  .course-section { padding: 48px 5% 55px; }
  .section-heading { align-items: flex-start; flex-direction: column; gap: 11px;
}
  .section-heading br { display: none; }
  .course-grid { grid-template-columns: repeat(2, minmax(0, 1fr)); gap: 14px; }
  .course-image-frame { aspect-ratio: .98; }
  .course-details { padding: 12px 11px 11px; }
  .course-details h3 { min-height: 48px; font-size: 17px; }
  .course-description { min-height: 68px; font-size: 9px; }
  .course-meta { font-size: 8px; }
  .course-choose { font-size: 9px; }
  .approach { padding: 34px 6%; grid-template-columns: 40px 1fr; gap: 14px; }
  .approach-index { font-size: 14px; }
  .approach h2 { font-size: 27px; }
  .approach-copy { grid-column: 2; }
  .registration { padding: 52px 6%; grid-template-columns: 1fr; gap: 27px; }
  .registration-copy h2 { font-size: 42px; }
  .footer { padding: 18px 5%; flex-wrap: wrap; }
  .footer > span, .footer > a:last-child { font-size: 8px; }
}

@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after { scroll-behavior: auto !important; transition-duration:
.01ms !important; }
}
```

ACADAMY.JS

```
const courseGrid = document.querySelector('#course-grid');
const courseSelect = document.querySelector('#course-id');
const registrationForm = document.querySelector('#registration-form');
const formStatus = document.querySelector('#form-status');
let availableCourses = [];

function createCourseCard(course) {
  const card = document.createElement('article');
  card.className = `course-card course-${course.color}`;
```

```
  const imageFrame = document.createElement('div');
  imageFrame.className = 'course-image-frame';
  const image = document.createElement('img');
  image.src = course.image;
  image.alt = course.alt;
  image.loading = 'lazy';
  const number = document.createElement('span');
  number.className = 'course-number';
  number.textContent = String(course.id).padStart(2, '0');
  imageFrame.append(image, number);

  const details = document.createElement('div');
  details.className = 'course-details';
  const category = document.createElement('p');
  category.className = 'course-category';
  category.textContent = course.category;
  const title = document.createElement('h3');
  title.textContent = course.title;
  const description = document.createElement('p');
  description.className = 'course-description';
  description.textContent = course.description;
  const meta = document.createElement('div');
  meta.className = 'course-meta';
  const duration = document.createElement('span');
  duration.textContent = course.duration;
  const level = document.createElement('span');
  level.textContent = course.level;
  meta.append(duration, level);
  const choose = document.createElement('button');
  choose.className = 'course-choose';
  choose.type = 'button';
  choose.dataset.courseId = String(course.id);
  choose.textContent = 'Choose this class';
  choose.setAttribute('aria-label', `Choose ${course.title}`);
  details.append(category, title, description, meta, choose);
  card.append(imageFrame, details);
  return card;
}

async function loadCourses() {
  try {
    const response = await fetch('/courses');
    if (!response.ok) throw new Error('Course list unavailable');
    availableCourses = await response.json();
```

```
    courseGrid.replaceChildren(...availableCourses.map(createCourseCard));
    for (const course of availableCourses) {
      const option = document.createElement('option');
      option.value = String(course.id);
      option.textContent = course.title;
      courseSelect.append(option);
    }
  } catch {
    const message = document.createElement('p');
    message.className = 'error-message';
    message.textContent = 'The course list could not load. Refresh the page to
try again.';
    courseGrid.replaceChildren(message);
    courseSelect.options[0].textContent = 'Courses are temporarily unavailable';
  }
}

courseGrid.addEventListener('click', (event) => {
  const button = event.target.closest('[data-course-id]');
  if (!button ||
!courseSelect.querySelector(`option[value="${button.dataset.courseId}"]`))
return;
  courseSelect.value = button.dataset.courseId;
  document.querySelector('#register').scrollIntoView({ behavior: 'smooth' });
  courseSelect.focus({ preventScroll: true });
});

registrationForm.addEventListener('submit', async (event) => {
  event.preventDefault();
  formStatus.textContent = '';
  formStatus.classList.remove('is-error');

  if (!registrationForm.reportValidity()) return;
  const selectedCourse = availableCourses.find((course) => String(course.id) ===
courseSelect.value);
  if (!selectedCourse) {
    courseSelect.setCustomValidity('Choose one of the available courses.');
    courseSelect.reportValidity();
    courseSelect.setCustomValidity('');
    return;
  }

  const submitButton = registrationForm.querySelector('button[type="submit"]');
```

```
  submitButton.disabled = true;
  submitButton.setAttribute('aria-busy', 'true');
  formStatus.textContent = 'Sending your request...';

  try {
    const response = await fetch('/registrations', {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(Object.fromEntries(new FormData(registrationForm))),
    });
    const result = await response.json();
    if (!response.ok) throw new Error(result.error || 'Your request could not be
sent. Please try again.');
    formStatus.textContent = result.message;
    registrationForm.reset();
  } catch (error) {
    formStatus.textContent = error.message || 'Something went wrong. Please try
again.';
    formStatus.classList.add('is-error');
  } finally {
    submitButton.disabled = false;
    submitButton.removeAttribute('aria-busy');
  }
});

loadCourses();
```

ACADEMY.HTML

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <meta name="theme-color" content="#f4f5ee">
  <meta name="description" content="Short, human-sized courses for curious
people. Find your next thing at Fieldnote Study Club.">
  <title>Fieldnote Study Club | Keep learning interesting</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
href="https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;600;700&family
```

```
=Fraunces:opsz,wght@9..144,500;9..144,600;9..144,700&display=swap"
rel="stylesheet">
  <link rel="stylesheet" href="{{ url_for('static', filename='academy.css') }}">
  <script src="{{ url_for('static', filename='academy.js') }}" defer></script>
</head>
<body>
  <div class="topline"><span>FIELDNOTE STUDY CLUB</span><span>Small classes for
curious people <b aria-hidden="true">✳</b> New term begins October
26</span></div>
  <header class="site-header">
    <a class="brand" href="#home" aria-label="Fieldnote Study Club home"><span
class="brand-symbol" aria-hidden="true">f.</span><span>fieldnote<small>STUDY
CLUB</small></span></a>
    <nav class="navigation" aria-label="Main navigation"><a
href="#courses">Explore courses</a><a href="#approach">Our approach</a><a
class="nav-cta" href="#register">Find your class <span aria-
hidden="true">↗</span></a></nav>
  </header>

  <main id="home">
    <section class="hero" aria-labelledby="hero-title">
      <div class="hero-copy"><p class="eyebrow"><span class="eyebrow-
line"></span> A little room to grow</p><h1 id="hero-title">Make room<br>for
<em>what’s next.</em></h1><p class="hero-description">Good classes make the world
feel bigger. Meet a new idea, practice a new skill, and see where it takes
you.</p><div class="hero-actions"><a class="button button-ink"
href="#courses">See the courses <span aria-hidden="true">↓</span></a><span
class="class-note"><b>06</b> small-group classes</span></div><div class="hero-
footnote">No prerequisites. Just bring your curiosity.</div></div>
      <div class="hero-visual"><img src="https://images.unsplash.com/photo-
1523580846011-d3a5bc25702b?auto=format&fit=crop&w=1500&q=90" alt="Students
sharing ideas around a classroom table" fetchpriority="high"><div class="hero-
sticker"><span>LEARN</span><strong>at your<br>own pace</strong><span>FIELDNOTE ·
2026</span></div><p class="image-caption">A good question can take you
anywhere.</p></div>
    </section>

    <section class="course-section" id="courses" aria-labelledby="courses-title">
      <div class="section-heading"><div><p class="eyebrow">A fresh page, every
term</p><h2 id="courses-title">Pick up <em>something new.</em></h2></div><p>Short
courses. Thoughtful teachers.<br>Room to try things out.</p></div>
      <div class="course-grid" id="course-grid" aria-live="polite"><p
class="loading-message">Gathering the course list...</p></div>
```

```
    </section>

    <section class="approach" id="approach" aria-labelledby="approach-title"><div
class="approach-index">01 <span>/ 03</span></div><div><p
class="eyebrow">Learning, made human</p><h2 id="approach-title">A little
structure.<br><em>A lot of possibility.</em></h2></div><p class="approach-
copy">Fieldnote is a small learning club for people who like following a
question. Join a live cohort, learn alongside other beginners, and leave with a
skill you can use beyond the classroom.</p></section>

    <section class="registration" id="register" aria-labelledby="register-
title"><div class="registration-copy"><p class="eyebrow">Your next chapter starts
here</p><h2 id="register-title">Save your <em>seat.</em></h2><p>Tell us where to
reach you and choose the class you'd like to join. We'll send the details by
email.</p><div class="registration-aside"><span class="aside-star" aria-
hidden="true">✳</span><span>Small cohorts, so every question has
room.</span></div></div>
      <form class="registration-form" id="registration-form" novalidate>
        <label for="full-name">Your name</label><input id="full-name"
name="full_name" autocomplete="name" required minlength="2" maxlength="80"
placeholder="First and last name">
        <label for="email">Email address</label><input id="email" name="email"
type="email" autocomplete="email" required placeholder="you@example.com">
        <label for="course-id">Choose a course</label><select id="course-id"
name="course_id" required><option value="">Select a class</option></select>
        <div class="form-bottom"><p class="form-status" id="form-status"
role="status" aria-live="polite"></p><button class="button button-coral"
type="submit">Request a place <span aria-hidden="true">↗</span></button></div>
      </form>
    </section>
  </main>

  <footer class="footer"><a class="brand" href="#home"><span class="brand-symbol"
aria-hidden="true">f.</span><span>fieldnote<small>STUDY
CLUB</small></span></a><span>Stay curious. Take notes.</span><a
href="mailto:hello@fieldnotestudy.club">hello@fieldnotestudy.club</a></footer>
</body>
</html>
```

ACADEMY_APP.PY

```
import re
```

```
from flask import Flask, jsonify, render_template, request

app = Flask(__name__)

COURSES = [
    {
        "id": 1,
        "title": "Creative Coding with Python",
        "category": "Technology",
        "duration": "6 weeks",
        "level": "Beginner",
        "description": "Build small, expressive programs while learning the
foundations of Python.",
        "image": "https://images.unsplash.com/photo-1516321318423-
f06f85e504b3?auto=format&fit=crop&w=900&q=85",
        "alt": "Student learning on a laptop",
        "color": "lime",
    },
    {
        "id": 2,
        "title": "Visual Storytelling",
        "category": "Design",
        "duration": "4 weeks",
        "level": "All levels",
        "description": "Turn everyday observations into thoughtful photographs
and short stories.",
        "image": "https://images.unsplash.com/photo-1455390582262-
044cdea277a8?auto=format&fit=crop&w=900&q=85",
        "alt": "Notebook and pen ready for a writing session",
        "color": "blue",
    },
    {
        "id": 3,
        "title": "The Psychology of Habits",
        "category": "Personal growth",
        "duration": "3 weeks",
        "level": "All levels",
        "description": "Understand how routines take shape and make change easier
to sustain.",
        "image": "https://images.unsplash.com/photo-1506784983877-
45594efa4cbe?auto=format&fit=crop&w=900&q=85",
        "alt": "Open journal for reflecting on new habits",
        "color": "coral",
```

```
    },
    {
        "id": 4,
        "title": "Designing for the Web",
        "category": "Design",
        "duration": "8 weeks",
        "level": "Intermediate",
        "description": "Create useful, accessible digital experiences from sketch
to prototype.",
        "image": "https://images.unsplash.com/photo-1498050108023-
c5249f4df085?auto=format&fit=crop&w=900&q=85",
        "alt": "Web designer working at a laptop",
        "color": "blue",
    },
    {
        "id": 5,
        "title": "Everyday Data Skills",
        "category": "Technology",
        "duration": "5 weeks",
        "level": "Beginner",
        "description": "Find the useful story in a spreadsheet and communicate it
clearly.",
        "image": "https://images.unsplash.com/photo-1551288049-
bebda4e38f71?auto=format&fit=crop&w=900&q=85",
        "alt": "Colorful data charts on a screen",
        "color": "lime",
    },
    {
        "id": 6,
        "title": "Writing for Change",
        "category": "Communication",
        "duration": "4 weeks",
        "level": "All levels",
        "description": "Shape a clear message that helps people understand and
take action.",
        "image": "https://images.unsplash.com/photo-1456324504439-
367cee3b3c32?auto=format&fit=crop&w=900&q=85",
        "alt": "Writer planning ideas in a notebook",
        "color": "coral",
    },
]
```

```
# Registrations are held in memory for this demo and reset when the server
restarts.
REGISTRATIONS = []
EMAIL_PATTERN = re.compile(r"^[^\s@]+@[^\s@]+\.[^\s@]+$")



@app.get("/")
def home():
    return render_template("academy.html")



@app.get("/courses")
def get_courses():
    return jsonify(COURSES)



@app.post("/registrations")
def create_registration():
    payload = request.get_json(silent=True)
    if not isinstance(payload, dict):
        return jsonify({"error": "Please submit registration details as JSON."}),
400

    full_name = payload.get("full_name")
    email = payload.get("email")
    course_id = payload.get("course_id")
    if not isinstance(full_name, str) or not isinstance(email, str) or course_id
is None:
        return jsonify({"error": "Name, email, and course are required."}), 400

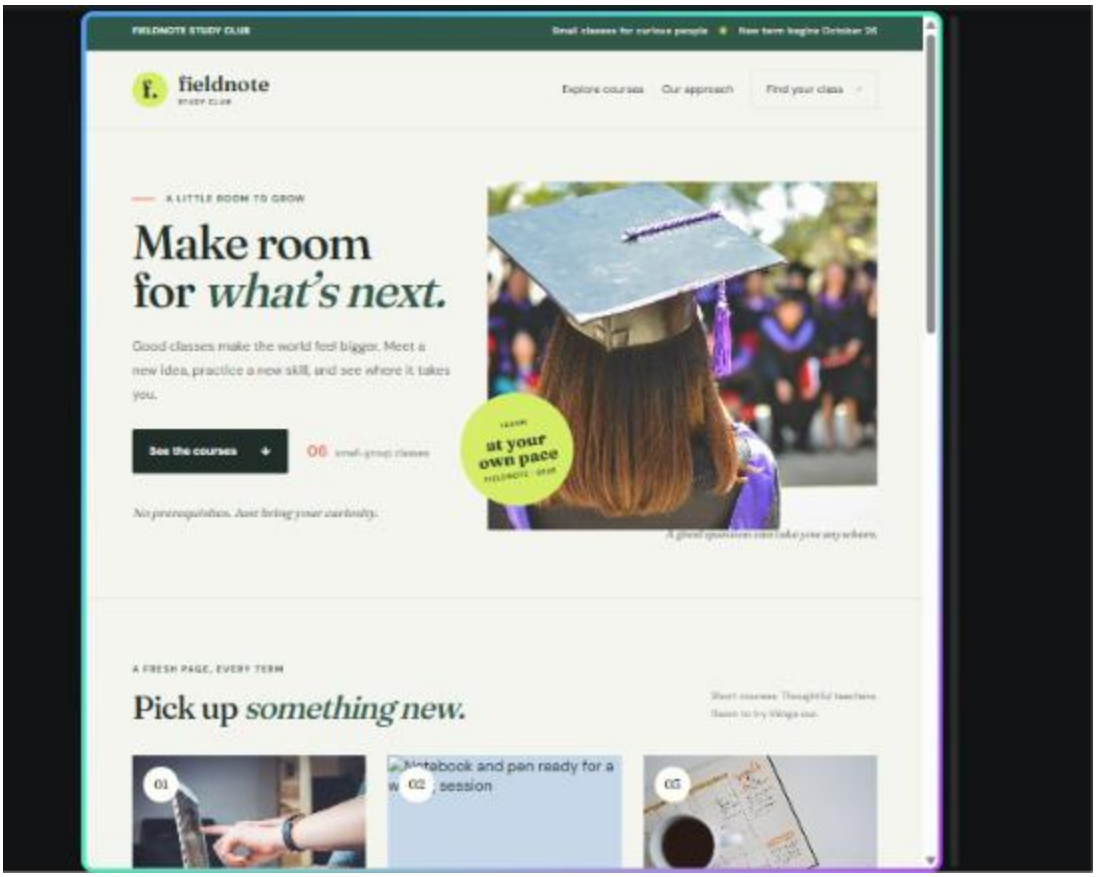
    full_name = full_name.strip()
    email = email.strip()
    if len(full_name) < 2 or len(full_name) > 80:
        return jsonify({"error": "Name must be between 2 and 80 characters."}),
400
    if not EMAIL_PATTERN.fullmatch(email):
        return jsonify({"error": "Please provide a valid email address."}), 400
    if isinstance(course_id, bool) or not isinstance(course_id, (str, int)):
        return jsonify({"error": "Please select an available course."}), 400
    try:
        course_id = int(course_id)
    except ValueError:
        return jsonify({"error": "Please select an available course."}), 400
```

```
    course = next((item for item in COURSES if item["id"] == course_id), None)
    if course is None:
        return jsonify({"error": "Please select an available course."}), 400

    registration = {
        "id": len(REGISTRATIONS) + 1,
        "full_name": full_name,
        "email": email,
        "course_id": course["id"],
        "course_title": course["title"],
    }
    REGISTRATIONS.append(registration)
    app.logger.info("New registration for %s: %s <%s>", course["title"],
full_name, email)
    return jsonify({"message": "Your place is requested. We will be in touch
soon.", "registration": registration}), 201


if __name__ == "__main__":
    app.run(debug=True, port=5001)
```

OUTPUT


fieldnote
Explore courses
Our approach
Find your class
A LITTLE ROOM TO GROW
Make room for what's next.
Good classes make the world feel bigger. Meet a new idea, practice a new skill, and see where it takes you.
See the courses
06
No prerequisites. Just bring your curiosity.
at your own pace
A FRESH PAGE, EVERY TERM
Pick up something new.
01
02
session
03

Set 3: Online Bookstore Website

Story: A small bookstore wants a website to display available books and allow customers to search for books by title.

Task:

1. Use AI to generate a responsive website with book images, titles, prices, a search bar, and JavaScript search functionality.
2. Use AI to create REST API endpoints: GET /books to retrieve books and GET /books/{id} to retrieve a specific book.
3. Test the website and API to verify book listings, search results, and individual book details.

PROMPT

- Create a responsive website displaying book images, titles, authors, and prices.
- Add a search bar that searches books by title using JavaScript.
- Create Flask REST API endpoints:
  - `GET /books` — retrieve all books.
  - `GET /books/<id>` — retrieve an individual book by ID.
- Test book listings, search results, and individual book details.

CODE

BOOKS.CSS

```
@import
url('https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;600;700&family=
Fraunces:opsz,wght@9..144,500;9..144,600;9..144,700&display=swap');

:root {
  color-scheme: light;
  --paper: #f0ede5;
  --white: #fbfaf6;
  --ink: #292f2b;
  --muted: #797b72;
  --green: #355847;
```

```
  --orange: #c66343;
  --blue: #a7bec4;
  --yellow: #e6c75d;
  --line: #d8d3c8;
  --serif: 'Fraunces', Georgia, serif;
  --sans: 'DM Sans', sans-serif;
}

* { box-sizing: border-box; }
html { scroll-behavior: smooth; }
body { margin: 0; background: var(--paper); color: var(--ink); font: 14px var(--
sans); }
a { color: inherit; }
.shop-note { min-height: 32px; padding: 8px 5.5%; display: flex; justify-content:
space-between; gap: 15px; background: var(--green); color: #f7f4eb; font-size:
8px; letter-spacing: .08em; }
.shop-note > span:first-child { font-weight: 700; }
.shop-note i { margin-left: 7px; color: var(--yellow); font-style: normal; }
.header { min-height: 76px; padding: 12px 5.5%; display: flex; justify-content:
space-between; align-items: center; border-bottom: 1px solid var(--line); }
.brand { font: 600 25px/.9 var(--serif); letter-spacing: -.04em; text-decoration:
none; }
.brand span { font-weight: 500; }
.brand b { color: var(--orange); font-family: var(--sans); }
.header nav { display: flex; align-items: center; gap: 31px; }
.header nav a { font-size: 10px; text-decoration: none; }
.header nav a:hover, .footer > a:last-child:hover { color: var(--orange); }
.header nav .header-link { padding: 10px 13px; border: 1px solid var(--line);
border-radius: 2px; }
.header-link span { margin-left: 8px; color: var(--orange); }
.intro { padding: 51px 5.5% 66px; display: grid; grid-template-columns: .83fr
1.17fr; gap: 6.5%; align-items: center; }
.intro-copy { padding-left: 5%; }
.eyebrow { margin: 0 0 15px; color: var(--green); font-size: 9px; font-weight:
700; letter-spacing: .14em; text-transform: uppercase; }
.intro-copy .eyebrow span { margin-right: 7px; color: var(--orange); }
h1, h2 { margin: 0; font-family: var(--serif); font-weight: 500; letter-spacing:
-.035em; }
h1 { font-size: clamp(47px, 5.2vw, 73px); line-height: .98; }
h1 em, h2 em { color: var(--orange); font-weight: 500; }
.intro-description { max-width: 360px; margin: 20px 0 23px; color: #65695f; font-
size: 12px; line-height: 1.85; }
.button { min-height: 43px; padding: 0 16px; display: inline-flex; justify-
content: center; align-items: center; gap: 25px; border-radius: 2px; background:
```

```
var(--ink); color: #fff; font-size: 10px; font-weight: 700; text-decoration:
none; transition: transform .18s ease, background .18s ease; }
.button:hover { transform: translateY(-2px); background: var(--green); }
.intro-aside { margin: 26px 0 0; color: var(--muted); font: italic 11px var(--
serif); }
.intro-image { min-height: 385px; position: relative; }
.intro-image > img { width: 100%; height: 410px; display: block; object-fit:
cover; object-position: center 52%; }
.image-note { width: 108px; height: 108px; position: absolute; left: -26px;
bottom: 25px; display: flex; flex-direction: column; justify-content: center;
align-items: center; gap: 4px; border-radius: 50%; background: var(--yellow);
text-align: center; transform: rotate(-8deg); }
.image-note span { font-size: 6px; font-weight: 700; letter-spacing: .12em; }
.image-note strong { font: 600 17px/.95 var(--serif); }
.image-caption { position: absolute; right: 0; bottom: -22px; color: var(--
muted); font: italic 11px var(--serif); }
.collection { padding: 62px 5.5% 75px; border-top: 1px solid var(--line); }
.collection-heading { display: flex; justify-content: space-between; align-items:
end; gap: 24px; }
h2 { font-size: clamp(32px, 3.3vw, 44px); line-height: 1.1; }
.search-wrap { width: min(100%, 310px); min-height: 43px; padding: 0 12px;
display: flex; align-items: center; gap: 10px; border: 1px solid #c8c5ba;
background: var(--white); }
.search-icon { width: 14px; height: 14px; position: relative; flex: 0 0 14px;
border: 1.5px solid var(--green); border-radius: 50%; }
.search-icon::after { width: 6px; height: 1.5px; position: absolute; right: -4px;
bottom: -2px; background: var(--green); content: ''; transform: rotate(48deg); }
.search-wrap input { width: 100%; min-width: 0; border: 0; outline: 0;
background: transparent; color: var(--ink); font: 11px var(--sans); }
.search-wrap input::placeholder { color: #818279; }
.search-wrap:focus-within { border-color: var(--green); box-shadow: 0 0 0 2px
#35584720; }
.search-hint { color: #929287; font-size: 7px; font-weight: 700; letter-spacing:
.1em; }
.results-bar { margin: 20px 0 15px; padding-bottom: 10px; display: flex; justify-
content: space-between; border-bottom: 1px solid var(--line); color: var(--
muted); font-size: 9px; }
.book-grid { display: grid; grid-template-columns: repeat(4, minmax(0, 1fr));
gap: 30px 22px; }
.book-card { min-width: 0; }
.cover-frame { aspect-ratio: .73; overflow: hidden; background: #ddd8cb; }
.cover-frame img { width: 100%; height: 100%; display: block; object-fit: cover;
transition: transform .35s ease; }
.book-card:hover .cover-frame img { transform: scale(1.035); }
```

```
.book-genre { margin: 13px 0 5px; color: var(--green); font-size: 8px; font-
weight: 700; letter-spacing: .1em; text-transform: uppercase; }
.book-card h3 { margin: 0; font: 600 19px/1.18 var(--serif); }
.book-author { margin: 5px 0 10px; color: var(--muted); font-size: 10px; }
.book-card-footer { display: flex; justify-content: space-between; align-items:
center; gap: 9px; }
.book-price { color: var(--orange); font-size: 11px; font-weight: 700; }
.details-button { padding: 6px 0 5px 10px; border: 0; background: transparent;
color: var(--ink); font: 600 9px var(--sans); cursor: pointer; }
.details-button::after { margin-left: 7px; color: var(--orange); content: '↗'; }
.details-button:hover { color: var(--green); }
.empty-state { grid-column: 1 / -1; padding: 35px 0; color: var(--muted); font:
17px var(--serif); }
.about { padding: 43px 8%; display: grid; grid-template-columns: 80px 1fr 1fr;
gap: 28px; align-items: center; background: #e4e1d6; }
.about-mark { width: 66px; height: 66px; display: grid; place-items: center;
border: 1px solid #a9aa98; border-radius: 50%; color: var(--orange); font: 25px
var(--serif); }
.about-mark span { margin: 20px 0 0 -21px; color: var(--green); }
.about h2 { font-size: 32px; }
.about > p { max-width: 370px; margin: 0; color: #63675e; font-size: 10px; line-
height: 1.8; }
.book-dialog { width: min(620px, calc(100% - 32px)); max-width: 620px; padding:
24px; position: fixed; inset: 0; border: 1px solid var(--line); background: var(-
-paper); color: var(--ink); box-shadow: 0 20px 80px #141c1640; }
.book-dialog::backdrop { background: #17221e99; }
.book-dialog[open] { display: grid; grid-template-columns: minmax(130px, .7fr)
1fr; align-items: center; gap: 25px; }
.dialog-close { width: 32px; height: 32px; position: absolute; top: 10px; right:
10px; display: grid; place-items: center; border: 1px solid var(--line); border-
radius: 50%; background: var(--white); color: var(--ink); font: 20px var(--sans);
cursor: pointer; }
.dialog-cover { width: 100%; max-height: 340px; object-fit: cover; }
.dialog-copy { padding: 16px 10px 16px 0; }
.dialog-copy h2 { padding-right: 14px; font-size: 32px; }
.dialog-author { margin: 8px 0 18px; color: var(--muted); font-size: 11px; }
.dialog-description { color: #65695f; font-size: 11px; line-height: 1.8; }
.dialog-price { margin: 20px 0 0; color: var(--orange); font-weight: 700; }
.footer { min-height: 74px; padding: 15px 5.5%; display: flex; justify-content:
space-between; align-items: center; gap: 16px; border-top: 1px solid var(--line);
}
.footer .brand { font-size: 19px; }
.footer > span, .footer > a:last-child { color: var(--muted); font-size: 9px; }
```

```css
@media (max-width: 900px) {
  .header nav { gap: 19px; }
  .intro { grid-template-columns: .9fr 1.1fr; gap: 5%; }
  .intro-copy { padding-left: 0; }
  .intro-image, .intro-image > img { min-height: 340px; height: 340px; }
  .book-grid { grid-template-columns: repeat(3, minmax(0, 1fr)); }
  .about { padding: 40px 6%; grid-template-columns: 65px 1fr 1fr; gap: 19px; }
}

@media (max-width: 620px) {
  .shop-note { padding: 8px 5%; font-size: 7px; }
  .shop-note > span:first-child { display: none; }
  .shop-note > span:last-child { margin: auto; }
  .header { min-height: 70px; padding: 12px 5%; align-items: flex-start; flex-
direction: column; gap: 12px; }
  .header nav { width: 100%; justify-content: space-between; gap: 8px; }
  .header nav a { font-size: 9px; }
  .header nav .header-link { padding: 8px; }
  .intro { padding: 39px 5% 49px; grid-template-columns: 1fr; gap: 26px; }
  h1 { font-size: 56px; }
  .intro-description { max-width: 420px; }
  .intro-image, .intro-image > img { min-height: 280px; height: 280px; }
  .image-note { width: 91px; height: 91px; left: -8px; bottom: 19px; }
  .image-note strong { font-size: 15px; }
  .collection { padding: 47px 5% 54px; }
  .collection-heading { align-items: flex-start; flex-direction: column; gap:
17px; }
  .search-wrap { width: 100%; }
  .book-grid { grid-template-columns: repeat(2, minmax(0, 1fr)); gap: 25px 13px;
}
  .book-genre { margin-top: 9px; font-size: 7px; }
  .book-card h3 { font-size: 17px; }
  .book-author { font-size: 9px; }
  .book-price { font-size: 10px; }
  .details-button { font-size: 8px; }
  .results-bar { margin-top: 16px; }
  .about { padding: 34px 6%; grid-template-columns: 54px 1fr; gap: 15px; }
  .about-mark { width: 50px; height: 50px; font-size: 20px; }
  .about-mark span { margin-top: 17px; margin-left: -17px; }
  .about h2 { font-size: 26px; }
  .about > p { grid-column: 2; }
  .book-dialog { padding: 18px; }
  .book-dialog[open] { grid-template-columns: minmax(95px, .65fr) 1fr; gap: 15px;
}
```

```
  .dialog-copy h2 { font-size: 25px; }
  .dialog-description { font-size: 10px; }
  .footer { padding: 18px 5%; flex-wrap: wrap; }
  .footer > span, .footer > a:last-child { font-size: 8px; }
}

@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after { scroll-behavior: auto !important; transition-duration:
.01ms !important; }
}
```

BOOKS.JS

```
const bookGrid = document.querySelector('#book-grid');
const bookSearch = document.querySelector('#book-search');
const resultCount = document.querySelector('#result-count');
const bookDialog = document.querySelector('#book-dialog');
let books = [];

function formatPrice(price) {
  return new Intl.NumberFormat('en-US', { style: 'currency', currency: 'USD'
}).format(price);
}

function createBookCard(book) {
  const card = document.createElement('article');
  card.className = 'book-card';
  const coverFrame = document.createElement('div');
  coverFrame.className = 'cover-frame';
  const cover = document.createElement('img');
  cover.src = book.cover;
  cover.alt = book.alt;
  cover.loading = 'lazy';
  coverFrame.append(cover);

  const genre = document.createElement('p');
  genre.className = 'book-genre';
  genre.textContent = book.genre;
  const title = document.createElement('h3');
  title.textContent = book.title;
  const author = document.createElement('p');
  author.className = 'book-author';
  author.textContent = book.author;
```

```
  const footer = document.createElement('div');
  footer.className = 'book-card-footer';
  const price = document.createElement('span');
  price.className = 'book-price';
  price.textContent = formatPrice(book.price);
  const details = document.createElement('button');
  details.className = 'details-button';
  details.type = 'button';
  details.dataset.bookId = String(book.id);
  details.textContent = 'Details';
  details.setAttribute('aria-label', `View details for ${book.title}`);
  footer.append(price, details);
  card.append(coverFrame, genre, title, author, footer);
  return card;
}

function renderBooks() {
  const query = bookSearch.value.trim().toLocaleLowerCase();
  const filteredBooks = books.filter((book) =>
book.title.toLocaleLowerCase().includes(query));
  resultCount.textContent = `${filteredBooks.length} ${filteredBooks.length === 1
? 'book' : 'books'}`;
  if (filteredBooks.length === 0) {
    const empty = document.createElement('p');
    empty.className = 'empty-state';
    empty.textContent = 'No titles match that search. Try another title.';
    bookGrid.replaceChildren(empty);
    return;
  }
  bookGrid.replaceChildren(...filteredBooks.map(createBookCard));
}

async function loadBooks() {
  try {
    const response = await fetch('/books');
    if (!response.ok) throw new Error('Books could not be loaded.');
    books = await response.json();
    renderBooks();
  } catch {
    resultCount.textContent = 'Shelf unavailable';
    const error = document.createElement('p');
    error.className = 'empty-state';
    error.textContent = 'The books could not load. Please refresh to try again.';
    bookGrid.replaceChildren(error);
```

```
  }
}

bookSearch.addEventListener('input', renderBooks);

bookGrid.addEventListener('click', async (event) => {
  const button = event.target.closest('[data-book-id]');
  if (!button) return;
  button.disabled = true;
  try {
    const response = await fetch(`/books/${button.dataset.bookId}`);
    const book = await response.json();
    if (!response.ok) throw new Error(book.error || 'Book details unavailable.');
    document.querySelector('#dialog-cover').src = book.cover;
    document.querySelector('#dialog-cover').alt = book.alt;
    document.querySelector('#dialog-genre').textContent = book.genre;
    document.querySelector('#dialog-title').textContent = book.title;
    document.querySelector('#dialog-author').textContent = `by ${book.author}`;
    document.querySelector('#dialog-description').textContent = book.description;
    document.querySelector('#dialog-price').textContent =
formatPrice(book.price);
    bookDialog.showModal();
  } catch {
    button.textContent = 'Unavailable';
  } finally {
    button.disabled = false;
  }
});

document.querySelector('.dialog-close').addEventListener('click', () =>
bookDialog.close());
bookDialog.addEventListener('click', (event) => {
  if (event.target === bookDialog) bookDialog.close();
});

loadBooks();
```

BOOKS.HTML

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <meta name="theme-color" content="#f0ede5">
  <meta name="description" content="A curious little bookshop for good stories,
big ideas, and the books you keep close.">
  <title>Margin Notes | Books for the in-between</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
href="https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;600;700&family
=Fraunces:opsz,wght@9..144,500;9..144,600;9..144,700&display=swap"
rel="stylesheet">
  <link rel="stylesheet" href="{{ url_for('static', filename='books.css') }}">
  <script src="{{ url_for('static', filename='books.js') }}" defer></script>
</head>
<body>
  <div class="shop-note"><span>MARGIN NOTES BOOKSHOP</span><span>Independent
books for curious minds <i aria-hidden="true">✳</i> Free local
pickup</span></div>
  <header class="header">
    <a class="brand" href="#home" aria-label="Margin Notes
home">margin<span>notes</span><b>.</b></a>
    <nav aria-label="Main navigation"><a href="#collection">The collection</a><a
href="#about">Our point of view</a><a class="header-link"
href="#collection">Browse books <span aria-hidden="true">↘</span></a></nav>
  </header>

  <main id="home">
    <section class="intro" aria-labelledby="intro-title">
      <div class="intro-copy"><p class="eyebrow"><span aria-
hidden="true">✳</span> A bookshop for the in-between</p><h1 id="intro-
title">Find a story<br>that <em>stays with you.</em></h1><p class="intro-
description">For slow Sundays, long train rides, and the five minutes before bed
that turn into an hour. Come find your next favorite.</p><a class="button"
href="#collection">Explore the shelves <span aria-hidden="true">↓</span></a><p
class="intro-aside">Picked by people, not an algorithm.</p></div>
      <div class="intro-image"><img src="https://images.unsplash.com/photo-
1507842217343-583bb7270b66?auto=format&fit=crop&w=1500&q=90" alt="Sunlit shelves
lined with books" fetchpriority="high"><div class="image-note"><span>READ A
LITTLE</span><strong>stay<br>a while</strong><span>BOOKS OPEN
DOORS</span></div><span class="image-caption">A good book makes its own kind of
room.</span></div>
    </section>
```

```
    <section class="collection" id="collection" aria-labelledby="collection-
title">
      <div class="collection-heading"><div><p class="eyebrow">The current
shelf</p><h2 id="collection-title">Worth your <em>reading
time.</em></h2></div><label class="search-wrap" for="book-search"><span
class="search-icon" aria-hidden="true"></span><input id="book-search"
type="search" autocomplete="off" placeholder="Search titles" aria-label="Search
books by title"><span class="search-hint">TITLE</span></label></div>
      <div class="results-bar"><span id="result-count" aria-live="polite">Loading
books...</span><span>Eight hand-picked reads</span></div>
      <div class="book-grid" id="book-grid" aria-live="polite"><p class="loading-
message">Opening the shelves...</p></div>
    </section>

    <section class="about" id="about" aria-labelledby="about-title"><div
class="about-mark" aria-hidden="true">M<span>N</span></div><div><p
class="eyebrow">A small shop with a long list</p><h2 id="about-title">Books for
living<br><em>with your eyes open.</em></h2></div><p>We choose books that invite
a second look: clear-eyed fiction, generous ideas, and voices you'll want to hear
again. Browse at your own pace.</p></section>
  </main>

  <dialog class="book-dialog" id="book-dialog" aria-labelledby="dialog-
title"><button class="dialog-close" type="button" aria-label="Close book
details">×</button><img class="dialog-cover" id="dialog-cover" alt=""><div
class="dialog-copy"><p class="eyebrow" id="dialog-genre"></p><h2 id="dialog-
title"></h2><p class="dialog-author" id="dialog-author"></p><p class="dialog-
description" id="dialog-description"></p><p class="dialog-price" id="dialog-
price"></p></div></dialog>
  <footer class="footer"><a class="brand"
href="#home">margin<span>notes</span><b>.</b></a><span>Good books. Better
company.</span><a
href="mailto:hello@marginnotes.books">hello@marginnotes.books</a></footer>
</body>
</html>
```

BOOKS_APP.PY

```
from flask import Flask, jsonify, render_template

app = Flask(__name__)

BOOKS = [
```

```
    {
        "id": 1,
        "title": "The Creative Act",
        "author": "Rick Rubin",
        "price": 24.00,
        "genre": "Creativity",
        "description": "A generous guide to creative practice, attention, and
bringing ideas into the world.",
        "cover": "https://images.unsplash.com/photo-1544947950-
fa07a98d237f?auto=format&fit=crop&w=800&q=85",
        "alt": "A stack of books in warm natural light",
    },
    {
        "id": 2,
        "title": "Braiding Sweetgrass",
        "author": "Robin Wall Kimmerer",
        "price": 18.50,
        "genre": "Nature & Culture",
        "description": "Indigenous wisdom, botanical knowledge, and the lessons
plants offer about reciprocity.",
        "cover": "https://images.unsplash.com/photo-1512820790803-
83ca734da794?auto=format&fit=crop&w=800&q=85",
        "alt": "Open book resting on a wooden table",
    },
    {
        "id": 3,
        "title": "Small Things Like These",
        "author": "Claire Keegan",
        "price": 16.00,
        "genre": "Fiction",
        "description": "A quiet, beautifully observed story about conscience,
courage, and a winter in Ireland.",
        "cover": "https://images.unsplash.com/photo-1519682337058-
a94d519337bc?auto=format&fit=crop&w=800&q=85",
        "alt": "A paperback book surrounded by soft winter light",
    },
    {
        "id": 4,
        "title": "Ways of Seeing",
        "author": "John Berger",
        "price": 17.00,
        "genre": "Art & Ideas",
```

```
        "description": "A sharp, inviting exploration of how images shape what we notice and believe.",
        "cover": "https://images.unsplash.com/photo-1507842217343-583bb7270b66?auto=format&fit=crop&w=800&q=85",
        "alt": "Rows of books on a library shelf",
    },
    {
        "id": 5,
        "title": "Crying in H Mart",
        "author": "Michelle Zauner",
        "price": 19.00,
        "genre": "Memoir",
        "description": "A tender memoir of family, food, grief, and finding your way back to yourself.",
        "cover": "https://images.unsplash.com/photo-1543002588-bfa74002ed7e?auto=format&fit=crop&w=800&q=85",
        "alt": "A colorful book cover on a reading table",
    },
    {
        "id": 6,
        "title": "The Anthropocene Reviewed",
        "author": "John Green",
        "price": 20.00,
        "genre": "Essays",
        "description": "Thoughtful, funny essays on the small and enormous things that make up human life.",
        "cover": "https://images.unsplash.com/photo-1532012197267-da84d127e765?auto=format&fit=crop&w=800&q=85",
        "alt": "Books displayed upright on a shelf",
    },
    {
        "id": 7,
        "title": "A Psalm for the Wild-Built",
        "author": "Becky Chambers",
        "price": 15.50,
        "genre": "Science Fiction",
        "description": "A hopeful, gentle journey through questions of purpose, rest, and what people need.",
        "cover": "https://images.unsplash.com/photo-1513001900722-370f803f498d?auto=format&fit=crop&w=800&q=85",
        "alt": "A bright, illustrated-looking book on a desk",
    },
    {
```

```
        "id": 8,
        "title": "On Writing",
        "author": "Stephen King",
        "price": 18.00,
        "genre": "Writing",
        "description": "Part memoir and part practical advice on the craft and
habit of writing.",
        "cover": "https://images.unsplash.com/photo-1512820790803-
83ca734da794?auto=format&fit=crop&w=800&q=85",
        "alt": "Open book resting on a wooden table",
    },
]



@app.get("/")
def home():
    return render_template("books.html")



@app.get("/books")
def get_books():
    return jsonify(BOOKS)



@app.get("/books/<int:book_id>")
def get_book(book_id):
    book = next((item for item in BOOKS if item["id"] == book_id), None)
    if book is None:
        return jsonify({"error": "Book not found."}), 404
    return jsonify(book)



if __name__ == "__main__":
    app.run(debug=True, port=5002)
```

TEST_BOOKS.PY

```
import unittest

from books_app import app, BOOKS
```

```
class BookApiTests(unittest.TestCase):
    def setUp(self):
        self.client = app.test_client()

    def test_homepage_is_served(self):
        response = self.client.get("/")
        self.assertEqual(response.status_code, 200)
        self.assertIn(b"Margin Notes", response.data)

    def test_books_returns_full_catalog(self):
        response = self.client.get("/books")
        self.assertEqual(response.status_code, 200)
        books = response.get_json()
        self.assertEqual(len(books), len(BOOKS))
        self.assertTrue(all({"id", "title", "author", "price", "cover"} <=
book.keys() for book in books))

    def test_individual_book_returns_matching_details(self):
        expected = BOOKS[0]
        response = self.client.get(f"/books/{expected['id']}")
        self.assertEqual(response.status_code, 200)
        self.assertEqual(response.get_json(), expected)

    def test_unknown_book_returns_404(self):
        response = self.client.get("/books/999")
        self.assertEqual(response.status_code, 404)
        self.assertEqual(response.get_json()["error"], "Book not found.")



if __name__ == "__main__":
    unittest.main()
```

OUTPUT

http://127.0.0.1:5002/

A BOOKSHOP FOR THE IN-BETWEEN

Find a story that *stays with you.*

For slow Sundays, long train rides, and the five minutes before bed that turn into an hour. Come find your next favorite.

Explore the shelves ↓

Picked by people, not an algorithm.



A good book makes its own kind of room.

THE CURRENT SHELF

Worth your *reading time.*

Search titles TITLE

8 books

Eight hand-picked reads





